

Project Design Phase

Solution Architecture & Technical Design Document

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	3. Project Design Phase
Document Title	Solution Architecture Document
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Architecture Overview & Design Principles

The Solution Architecture for "Auto Ticket Classification using Flow Designer" establishes a robust, highly scalable, and secure technical blueprint. It adheres to key architectural principles: Native Cloud Execution, Separation of Concerns, Zero-Code Modularity, Data Normalization, and Comprehensive Auditability.

2. Multi-Tier Solution Architecture

Architectural Tier	ServiceNow Components & Artifacts	Responsibilities & Data Flows
Presentation & Interaction Tier	ServiceNow Form Designer / Form Builder Next Experience UI u_incident_workflow Form View	Captures user inputs (Caller, Short Description). Enforces mandatory field validation. Renders dynamic choice lists based on dictionary dependencies.
Process & Workflow Orchestration Tier	ServiceNow Workflow Studio / Flow Designer Flow: "Auto Classify School IT Tickets" Application Scope: Global	Listens for database record creation events. Evaluates conditional logic against incoming strings. Updates records asynchronously. Orchestrates outbound email communication.
Data Architecture & Persistence Tier	Custom Table: u_incident_workflow Auto-numbering Engine (Prefix: INC) sys_user, sys_user_group reference tables Choice List Dictionary Repository	Maintains persistent storage of incidents. Assigns auto-incrementing INC005xx keys. Enforces relational links to user and group entities. Validates dependent choice integrity.

Architectural Tier	ServiceNow Components & Artifacts	Responsibilities & Data Flows
Communication & Integration Tier	ServiceNow Outbound Email Engine sys_email Repository System Logs > Emails SMTP Gateway	Formats dynamic HTML email payloads. Resolves Caller email addresses via data pills. Dispatches outbound notifications. Records auditable delivery logs.
Lifecycle & Deployment Management	System Update Sets Local Update Set: "Project Update Set" XML Serialization / Deserialization Engine	Captures all table, dictionary, form design, and flow changes. Packages changes for promotion. Exports full solution as a portable XML artifact.

3. Data Model Schema: Table u_incident_workflow

The database layer is anchored by the custom table "u_incident_workflow". The following data dictionary defines the exact schema:

Field Label	Column Technical Name	Data Type	Reference / Choice Values	Attributes / Dependency
Number	u_number / sys_id	Auto Number	Prefix: INC Start: 500 Digits: 5	Read-Only, Unique Primary Key
Caller	u_caller	Reference	sys_user table	Mandatory, Reference Lookup
Category	u_category	Choice	Network, Hardware, Access, Performance	Dropdown with --None-- option
Subcategory	u_subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Compter	Dependent on: u_category
Short Description	u_short_description	String (255)	Free text entry	Mandatory, Evaluated by Flow
Description	u_description	String (4000)	Multiline detailed text	Optional supplemental context
State	u_state	Choice	New, In progress, On hold, Resolved, Closed	Default: New
Assigned Group	u_assigned_group	Reference	sys_user_group table	Assignment routing target
Assigned To	u_assigned_to	Reference	sys_user table	Specific technician assigned

4. Dynamic Choice Dependency Mapping Matrix

To ensure data integrity, Subcategory choices are strictly governed by their parent Category choice values as configured in the Dictionary Entry:

Parent Category Label	Category Backend Value	Dependent Subcategory Label	Subcategory Backend Value	Sequence
Network	network	Wi-Fi	wi-fi	1
Hardware	hardware	Projector	projector	1
Access	access	Forgot Password	forgot_password	1
Performance	performance	Slow Compter	slow_computer	1

5. Flow Designer Execution Logic Specification

The flow "Auto Classify School IT Tickets" operates according to the following sequential logic engine:

1. TRIGGER: Record Created on Table [u_incident_workflow] with Condition [Category is Empty].
2. BRANCH 1 (Wi-Fi): If Short Description contains "Wi-Fi" (or "Network")
--> Action: Update Record -> Category = Network, Subcategory = Wi-Fi
3. BRANCH 2 (Projector): Else If Short Description contains "Projector"
--> Action: Update Record -> Category = Hardware, Subcategory = Projector
4. BRANCH 3 (Password): Else If Short Description contains "Forgot password"
--> Action: Update Record -> Category = Access, Subcategory = Forgot Password
5. BRANCH 4 (Performance): Else If Short Description contains "Slow Computer"
--> Action: Update Record -> Category = Performance, Subcategory = Slow Compter
6. NOTIFICATION ACTION: Action "Send Email"
--> To: Trigger -> Incident Workflow Record -> Caller -> Email
--> Subject: "Your Request for the issue has been submitted."
--> Body: "Hi greetings from xyz college ...!!! We are viewed your request and solved by short time period. Thankyou. Ref:MSG..."